

Kaustav Banerjee

professor

基本信息

姓名: Kaustav Banerjee 所属院校: 加州大学-圣塔芭芭拉
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海外经历: 有

荣誉称号

★ APS Fellow

★ IEEE Fellow

★ AAAS Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

集成电路设计与集成系统

电子科学与技术

纳米材料与技术

研究领域

纳米电子与先进工艺

微电子器件与制造

微纳电子与纳米技术

低功耗电路与能量管理

半导体器件物理与工艺

纳米电子学与光子学

超大规模集成电路设计

低维材料与量子效应

EDA工具与电路验证

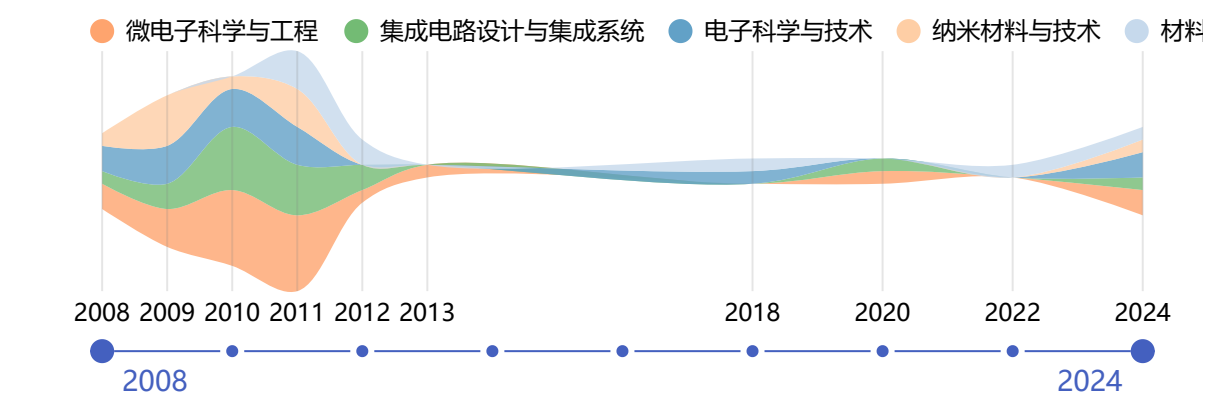
数字集成电路设计

个人简介

Kaustav Banerjee is a distinguished Professor in Electrical and Computer Engineering at the University of California, Santa Barbara. A world-renowned innovator in nanoelectronics, his research focuses on 2D materials, energy-efficient electronics, and photonics. Banerjee holds fellowships from

prestigious organizations including the American Association for the Advancement of Science and IEEE. He has received numerous honors, such as the IEEE Kiyo Tomiyasu Award and the Friedrich Wilhelm Bessel Research Award. Additionally, he is affiliated with several international institutions, including Tokyo Institute of Technology and Shanghai Jiao Tong University. Banerjee earned his PhD in Electrical Engineering and Computer Sciences from UC Berkeley.

近年发表的论文



Three-dimensional transistors with two-dimensional semiconductors for future CMOS scaling
NATURE ELECTRONICS
Pal, Arnab; Chavan, Tanmay; Jabbour, Jacob; Cao, Wei; Banerjee, Kaustav
被引用次数: 3 2024

An ultra energy-efficient hardware platform for neuromorphic computing enabled by 2D-TMD tunnel-FETs
NATURE COMMUNICATIONS
Pal, Arnab; Chai, Zichun; Jiang, Junkai; Cao, Wei; Davies, Mike; De, Vivek; Banerjee, Kaustav
被引用次数: 4 2024

Strain engineering in 2D FETs: Physics, status, and prospects
JOURNAL OF APPLIED PHYSICS
Kumar, Ankit; Xu, Lin; Pal, Arnab; Agashiwala, Kunjesh; Parto, Kamyar; Cao, Wei; Banerjee, Kaustav
被引用次数: 0 2024

Characterization and Closed-Form Modeling of Edge/Top/Hybrid Metal-2D Semiconductor Contacts
2022 INTERNATIONAL ELECTRON DEVICES MEETING, IEDM
Pal, Arnab; Mishra, Varun; Weber, Justin; Krishnaswamy, Karthik; Ghosh, Krishnendu; Penumatcha, Ashish Verma; Berrada, Salim; O'Brien, Kevin; Kencke, David; Banerjee, Kaustav

被引用次数: 0

2022

Is negative capacitance FET a steep-slope logic switch?

NATURE COMMUNICATIONS

Cao, Wei; Banerjee, Kaustav

被引用次数: 134

2020

Analytical Thermal Model for Self-Heating in Advanced FinFET Devices With Implications for Design and Reliability (vol 32, pg 1045, 2013)

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Xu, Chuan; Kolluri, Seshadri K.; Endo, Kazuhiko; Banerjee, Kaustav

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2020

On-chip intercalated-graphene inductors for next-generation radio frequency electronics

NATURE ELECTRONICS

Kang, Jiahao; Matsumoto, Yuji; Li, Xiang; Jiang, Junkai; Xie, Xuejun; Kawamoto, Keisuke; Kenmoku, Munehiro; Chu, Jae Hwan; Liu, Wei; Mao, Junfa; Ueno, Kazuyoshi; Banerjee, Kaustav

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2018

Subthreshold-swing physics of tunnel field-effect transistors

AIP ADVANCES

Cao, Wei; Sarkar, Deblina; Khatami, Yasin; Kang, Jiahao; Banerjee, Kaustav

被引用次数: 64

2014

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Xu, Chuan; Kolluri, Seshadri K.; Endo, Kazuhiko; Banerjee, Kaustav

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2013

Some Clarifications on Compact Modeling and Analysis of Through-Si-Via Induced Electrical Noise Coupling in Three-Dimensional ICs

IEEE TRANSACTIONS ON ELECTRON DEVICES

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被引用次数: 0

2012

Metal-to-Multilayer-Graphene Contact-Part II: Analysis of Contact Resistance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Khatami, Yasin; Li, Hong; Xu, Chuan; Banerjee, Kaustav

被引用次数: 36

2012

Metal-to-Multilayer-Graphene Contact-Part I: Contact Resistance Modeling

IEEE TRANSACTIONS ON ELECTRON DEVICES

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2012

Top illuminated inverted organic ultraviolet photosensors with single layer graphene electrodes

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Burkhardt, Martin; Liu, Wei; Shuttle, Christopher G.; Banerjee, Kaustav; Chabiny, Michael L.

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2012

Proposal for tunnel-field-effect-transistor as ultra-sensitive and label-free biosensors

APPLIED PHYSICS LETTERS

Sarkar, Deblina; Banerjee, Kaustav

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2012

Fast High-Frequency Impedance Extraction of Horizontal Interconnects and Inductors in 3-D ICs With Multiple Substrates

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

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High-Frequency Behavior of Graphene-Based Interconnects-Part I: Impedance Modeling

IEEE TRANSACTIONS ON ELECTRON DEVICES

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Metallic-nanoparticle assisted enhanced band-to-band tunneling current

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A Fully Analytical Model for the Series Impedance of Through-Silicon Vias With Consideration of Substrate Effects and Coupling With Horizontal Interconnects

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Compact Modeling and Analysis of Through-Si-Via-Induced Electrical Noise Coupling in Three-Dimensional ICs

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Xu, Chuan; Suaya, Roberto; Banerjee, Kaustav

被引用次数: 27

2011

Synthesis of high-quality monolayer and bilayer graphene on copper using chemical vapor deposition

CARBON

Liu, Wei; Li, Hong; Xu, Chuan; Khatami, Yasin; Banerjee, Kaustav

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2011

Vertically Stacked and Independently Controlled Twin-Gate MOSFETs on a Single Si Nanowire

IEEE ELECTRON DEVICE LETTERS

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2011

CMOS-Compatible Vertical-Silicon-Nanowire Gate-All-Around p-Type Tunneling FETs With ≤ 50 -mV/decade Subthreshold Swing

IEEE ELECTRON DEVICE LETTERS

Gandhi, Ramanathan; Chen, Zhixian; Singh, Navab; Banerjee, Kaustav; Lee, Sungjoo

被引用次数: 160

2011

A Physical Model for Work-Function Variation in Ultra-Short Channel Metal-Gate MOSFETs

IEEE ELECTRON DEVICE LETTERS

Rasouli, Seid Hadi; Xu, Chuan; Singh, Navab; Banerjee, Kaustav

被引用次数: 11

2011

Vertical Si-Nanowire n-Type Tunneling FETs With Low Subthreshold Swing (≤ 50 mV/decade) at Room Temperature

IEEE ELECTRON DEVICE LETTERS

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2011

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2010

Electron-hole duality during band-to-band tunneling process in graphene-nanoribbon tunnel-field-effect-transistors

APPLIED PHYSICS LETTERS

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2010

Analytical Expressions for High-Frequency VLSI Interconnect Impedance Extraction in the Presence of a Multilayer Conductive Substrate (vol 28, pg 1047, 2009)

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Design Optimization of FinFET Domino Logic Considering the Width Quantization Property

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Rasouli, Seid Hadi; Dadgour, Hamed F.; Endo, Kazuhiko; Koike, Hanpei; Banerjee, Kaustav

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Compact AC Modeling and Performance Analysis of Through-Silicon Vias in 3-D ICs

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A Novel Enhanced Electric-Field Impact-Ionization MOS Transistor

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A Novel Variation-Tolerant Keeper Architecture for High-Performance Low-Power Wide Fan-In Dynamic OR Gates

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Dadgour, Hamed F.; Banerjee, Kaustav

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Steep Subthreshold Slope n- and p-Type Tunnel-FET Devices for Low-Power and Energy-Efficient Digital Circuits

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High-Frequency Analysis of Carbon Nanotube Interconnects and Implications for On-Chip Inductor Design

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Li, Hong; Banerjee, Kaustav

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Carbon Nanomaterials for Next-Generation Interconnects and Passives: Physics, Status, and Prospects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Li, Hong; Xu, Chuan; Srivastava, Navin; Banerjee, Kaustav

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Modeling, Analysis, and Design of Graphene Nano-Ribbon Interconnects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Xu, Chuan; Li, Hong; Banerjee, Kaustav

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2009

On the Applicability of Single-Walled Carbon Nanotubes as VLSI Interconnects

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Srivastava, Navin; Li, Hong; Kreupl, Franz; Banerjee, Kaustav

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Circuit modeling and performance analysis of multi-walled carbon nanotube interconnects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Li, Hong; Yin, Wen-Yan; Banerjee, Kaustav; Mao, Jun-Fa

被引用次数: 299

2008

Cool chips: Opportunities and implications for power and thermal management

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lin, Sheng-Chih; Banerjee, Kaustav

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2008

Accurate Intrinsic Gate Capacitance Model for Carbon Nanotube-Array Based FETs Considering Screening Effect

IEEE ELECTRON DEVICE LETTERS

Kshirsagar, Chaitanya; Li, Hong; Kopley, Thomas E.; Banerjee, Kaustav

被引用次数: 32

2008

学术合作



Yasin Khatami	合作次数: 5
加州大学-圣塔芭芭拉	
毛军发	合作次数: 3
上海交通大学	
Kang, Jiahao	合作次数: 2
堪萨斯大学-劳伦斯	
Junkai Jiang	合作次数: 2
加州大学-圣塔芭芭拉	
Kunjesh Agashiwala	合作次数: 1
加州大学-圣塔芭芭拉	

还有 2 个合作者